Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 2: Combining Statements

Negation, Conjunction, Disjunction, and Truth Tables

Chartrand & Zhang, Section 1.2

Name: _____ Date: _____

Where we are

Monday we asked which sentences *are* statements. Today we take statements we already have and build new ones from them. Everything today rests on one question: if I know the truth values of the pieces, what is the truth value of the whole?

By the end of today you should be able to

1. Build a truth table with the correct number of rows, in the standard order.
2. Write the negation of a statement or an open sentence correctly.
3. Give the truth values of $P \wedge Q$, $P \vee Q$, and $P \oplus Q$ from those of P and Q.
4. Explain why "or" in mathematics always means *inclusive* or.
5. Build a truth table for a compound statement such as $(\sim P) \vee Q$.

Warm-up: the three sentences you brought

Trade papers with a neighbor. For each of their three sentences, decide: true statement, false statement, or not a statement. Then compare with what they intended.

Any disagreements? Those are the interesting ones.

1. Truth tables

Definition — truth table

A **truth table** is a table giving all possible assignments of truth values for the statements under consideration.

One statement P has 2 rows. Two statements P and Q have 4 rows. Three statements have 8 rows.

In general, n statements require 2^n rows — the same counting argument as Monday's password warm-up.

Use the book's standard order so your homework matches the text:

P	Q
T	T
T	F
F	T
F	F

What the four rows mean

Let P: *the tests pass*, and Q: *the build succeeds*. The four rows are four real situations:

(1) The tests pass and the build succeeds.

(2) The tests pass but the build does not succeed.

(3) The tests do not pass but the build succeeds.

(4) The tests do not pass and the build does not succeed.

A truth table is not an abstraction. It is a list of every case that could happen.

2. Negation

Definition 1.12 — negation

The **negation** of a statement P is the statement *not* P, written $\sim P$. Its truth value is always the opposite of P.

P	$\sim P$
T	F
F	T

Write the negation of each. Say it the way a person would actually say it.

P	$\sim P$
Broadway is a street in New York City.	Broadway is not a street in New York City.
The list is sorted.	The list is not sorted.
$x > 5$	$x \leq 5$
$(x-2)^2 > 0$	$(x-2)^2 \leq 0$

The two negation errors that cost the most points

1. Negation is not the opposite extreme. The negation of "the list is sorted" is "the list is not sorted" — *not* "the list is in reverse order." A list in random order is unsorted without being reversed.

2. The negation of $>$ is $\leq$, not $<$. If $x > 5$ is false, then x is 5 or below — and 5 itself is included. Dropping the equals sign is the single most common error in this section, and it is the same off-by-one that breaks loop bounds in code.

3. Conjunction

Definition 1.15 — conjunction

For statements P and Q, the **conjunction** of P and Q is the statement P *and* Q, written $P \wedge Q$. It is true **only** when P and Q are both true.

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

$P \wedge Q$ is true in 1 of the 4 rows.

"But" is just "and"

English has many words that carry contrast — *but, however, yet, although, while.* They differ in tone, not in truth conditions. All of them are $\wedge$.

"The tests pass *but* the build fails" and "the tests pass *and* the build fails" are the same statement logically. The first sounds surprised. Logic does not track surprise.

4. Disjunction — and the word that causes the most trouble

Definition 1.17 — disjunction

For statements P and Q, the **disjunction** of P and Q is the statement P *or* Q, written $P \vee Q$. It is false **only** when P and Q are both false — equivalently, it is true when *at least one* of them is true.

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

Look hard at the first row. If *both* are true, $P \vee Q$ is true.

In mathematics, "or" always includes "both"

Suppose you say: *"Before class tomorrow I'm stopping at the computer lab or going to the library."* You then go to both. Did you lie? **No.** You said at least one, and at least one happened.
Now suppose you have saved up and decide to buy *a Ford or a Toyota*. Here nobody expects you to buy both.
Ordinary English settles which reading applies using context. Mathematics cannot rely on context, so it *defines* the ambiguity away: $\vee$ is always inclusive. When we need the other meaning we use a different symbol entirely.

5. Exclusive or

Definition 1.18 — exclusive or

For statements P and Q, the **exclusive or** is the statement P *or* Q *but not both*, written $P \oplus Q$. It is true exactly when *exactly one* of P and Q is true.

P	Q	$P \vee Q$	$P \oplus Q$
T	T	T	F
T	F	T	T
F	T	T	T
F	F	F	F

The two columns differ in exactly one row — the row where both are true.

Note: $\oplus$ is rare in mathematics. Unless a problem says "but not both," assume $\vee$. It is common in computing, though — it is the ^ operator, and it is how parity bits and one-time pads work.

6. Building a bigger truth table

Method: give each piece its own column, work left to right, and combine only two columns at a time.

Build $(\sim P) \vee Q$.

P	Q	$\sim P$	$(\sim P) \vee Q$
T	T	F	T
T	F	F	F
F	T	T	T
F	F	T	T

This compound statement is false in 1 row: when P is true and Q is false.

Remember this table

$(\sim P) \vee Q$ turns out to be one of the most important compound statements in the course, for a reason that will not be obvious until Friday. Keep this page.

7. Discovery: is "not both" the same as "neither"?

Homework 1, Problem 5 asks whether a library policy matches a piece of code. Here is the same question stripped down. Fill in both tables and compare the last columns.

P	Q	$P \wedge Q$	$\sim(P \wedge Q)$	$(\sim P) \wedge (\sim Q)$
T	T	T	F	F
T	F	F	T	F
F	T	F	T	F
F	F	F	T	T

Do the last two columns agree? No

They differ in 2 rows. Which ones? TF and FT

Now try one more column. Add $(\sim P) \vee (\sim Q)$ and compare it with $\sim(P \wedge Q)$.

P	Q	$\sim(P \wedge Q)$	$(\sim P) \vee (\sim Q)$
T	T	F	F
T	F	T	T
F	T	T	T
F	F	T	T

What you just found

The negation of an *and* is an *or*, not an *and*. Negating a compound statement flips the connective as well as the pieces.
This has a name and a proof, and it is Theorem 1.24 in your book. We do it properly on Friday. For now: you have everything you need to finish Homework 1, Problem 5, and the two rows where the columns disagree are exactly the two cases where the library code misbehaves.

Symbols so far

Symbol	Name	True when	In code
$\sim P \quad (\neg P)$	negation	P is false	`!p  not p`
$P \wedge Q$	conjunction	both are true	`p && q  p and q`
$P \vee Q$	disjunction	at least one is true	`p || q  p or q`
$P \oplus Q$	exclusive or	exactly one is true	`p ^ q`

Exit ticket

1. P: "$x \geq 10$." Write $\sim P$.

2. Complete: $P \vee Q$ is false only when _____

3. Give one English sentence where "or" is inclusive, and one where it is exclusive.

Before Friday

Read Section 1.2 in full — pp. 13–24. You previewed pp. 13–15 for today.

Self-check (not collected). Section 1.2, Exercises 8(b), 8(c), and 9. Exercise 8(a) is the table we built in class, so use it to check your method before starting the others.

Homework 1 is due Friday, August 21, 11:59 p.m.

Submit one PDF to Canvas, and put the completed collaboration and AI statement first. Problem 5 is the library policy question — Section 7 above is the tool for it.

Questions I still have